Abdelrahman Ibrahim

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PROFILE

Experienced Principal Software Developer specializing in AUTOSAR stack configuration, ECU software integration, C/C++, and Python. Proven track record in memory stack management, safety module configuration, and diagnostic testing for automotive software solutions.

EDUCATION

B.Sc. Mechatronics Engineering Ain Shams University, Egypt Sep. 2013 – Jul. 2018

SKILLS

Programming:
- C/C++
- Python
- MATLAB

AUTOSAR Tools:
- DaVinci Developer & Configurator
- EB Tresos
- ISOLAR A/B
- AUTOSAR Builder

Testing & Debugging:
- CANoe
- VectorCAST
- Lauterbach TRACE32
- iSYSTEM

Process & VCS:
- JIRA
- Confluence
- Jazz
- Plastic SCM
- Git / GitHub

Microcontrollers:
- Renesas RH850
- Infineon AURIX TriCore
- NVIDIA Jetson TX2 / Nano

LANGUAGES

Arabic: Native
English: Advanced C1
German: A2

PROJECTS & EXPERIENCE

Principal Software Developer
Jul. 2026 – Present

Techno Welle GmbH, Germany

- Developed On-Screen Display (OSD) software components (SWCs).
- Developed video watermark detection features.
- Performed system debugging and defect resolution.
- Configured AUTOSAR RTE ports, runnables, DIDs, and NVRAM.
- Created debugger helper scripts and CANoe CAPL scripts for testing and validation purposes.

Vector DaVinci Developer & Configurator | IAR Debugger | Vector CANoe | Novatek T-CON Processors | Infineon AURIX TriCore | Infotainment ECU

Senior SW Engineer
Oct. 2025 – Jun. 2026

Techno Welle GmbH, Germany

- Configured AWS services including AWS IoT Core, DynamoDB, S3, EC2, and API Gateway for Fleet Marschal.
- Implemented a gateway to convert raw sensor data into readable formats.
- Developed the backend architecture for user, company, fleet, trip, and device management.
- Developed a simulator GUI using real-world data to test system reliability.
- Developed a performance analysis GUI to evaluate and compare metrics across different sensors.

AWS IoT Core | DynamoDB | S3 | EC2 | API Gateway | Python | AWS | Fleet Marschal | IOT

COURSES

- AWS Certified Machine Learning - Specialty
- Udacity Machine Learning Nanodegree
- Udacity Self-Driving Car Nanodegree
- Udacity Sensor Fusion Nanodegree
- Udacity C++ Nanodegree
- Udacity Computer Vision Nanodegree

Senior SW Engineer
May. 2023 – Aug. 2025

Techno Welle GmbH, Germany

- Implemented various Application Software Components (SWCs) including NVM, OBD, Histogram, and Passive Discharge.
- Configured modules across Memory, Communication, and Mode Management stacks.
- Configured ECU safety modules including FISTEST and ECM.
- Developed CAPL scripts and automated test modules to reproduce and analyze software bugs.
- Executed LDF and DEXT file integration for vehicle networking.

ETAS ISOLAR A/B | CANoe | iSYSTEM Debugger | Renesas RH850 | Compressor ECU

Senior SW Engineer
Oct. 2022 – Apr. 2023

Swift Act, Egypt

- Configured End-to-End (E2E) data protection mechanisms for CAN messages.
- Integrated CAN DBC files and Complex Device Drivers (CDDs).
- Analyzed software requirements and authored technical specification documents.

Vector DaVinci Developer & Configurator | Lauterbach TRACE32 | CANoe | Infineon AURIX TriCore | BMS ECU

SW Engineer
Jan. 2022 – Sep. 2022

ExpleoGroup, Egypt

- Configured AUTOSAR NVM (Non-Volatile Memory) BSW modules.
- Debugged auto-generated MATLAB/Simulink code and reported root-cause findings to application teams.
- Configured CAN signals, network interfaces, and message frames.

AUTOSAR Builder | EB Tresos | Lauterbach TRACE32 | CANoe | Infineon AURIX TriCore | Gateway ECU

Embedded SW Engineer
May. 2019 – Dec. 2021

Avelabs, Egypt

- Designed and configured NVM blocks using Vector DaVinci Developer and DaVinci Configurator.
- Implemented failure-detection indication flags for NVM read/write operations.
- Created Application Layer SWCs and configured RTE ports using Vector DaVinci Developer.
- Executed unit testing and bug fixing using VectorCAST, CANoe, and Lauterbach TRACE32.
- Managed issue tracking, branch merging workflows, and software releases using JIRA and Plastic SCM.

Vector DaVinci Developer & Configurator | VectorCAST | CANoe | Lauterbach TRACE32 | Infineon AURIX TriCore | ADAS ECU